

Lab Assignment 8

MTH 308: Numerical Analysis and Scientific Computing-I

Department of Mathematics and Statistics, IIT Kanpur

2026 Even Semester: April 10, 2026

This lab will have you implement the Lagrange form of the interpolating polynomial, the Newton form of the interpolating polynomial, and cubic splines to approximate functions. Please write a single script file named `Lab8LastnameRollNumber.m` and publish it as a PDF. Then submit the script `Lab8LastnameRollNumber.m` along with the corresponding published file `Lab8LastnameRollNumber.pdf` (for example, my submitted files would be `Lab8Biswas260102.m` and `Lab8Biswas260102.pdf`) to HELLO IITK.

Each solution should:

- Be contained in a separate cell, including the problem number and a short problem description.
- Be adequately commented to explain your logic.
- Run independently of other cells.

1. For $f(x) = \cos(\ln x)$, let $x_0 = 0.5$, $x_1 = 1$, $x_2 = 1.5$, and $x_3 = 2$. Construct Lagrange interpolating polynomials of degree at most one, at most two, and at most three (using appropriate nodes).

Plot the underlying function (as a solid black line), the interpolating polynomials (as dashed black lines), and the data points (as red circles) in a single figure window.

Approximate $f(1.75)$ using each interpolating polynomial and find the absolute error of each approximation. Which interpolating polynomial gives the best approximation for $f(1.75)$?

2. (a) Use divided differences to construct the interpolating polynomial of degree at most three for the following data:

x	$f(x)$
-0.10	17.3000
0.00	2.0000
0.20	5.1900
0.30	1.0000

- (b) Approximate $f(0.1)$ and $f(0.4)$ using the interpolating polynomial.
- (c) Plot the data (using red circles) and the interpolating polynomial (using a solid black line) for $x \in [-0.2, 0.5]$ in the same figure.
- (d) Add $f(0.05) = 3.1250$ to the data, construct the interpolating polynomial of degree at most four, and approximate $f(0.1)$ and $f(0.4)$.
- (e) Create a new plot that includes the data (using red circles), the interpolating polynomial from (a) (using a solid black line), and the interpolating polynomial from (d) (using a solid blue line).

Note: It would be useful to implement divided differences with a function that accepts vectors for the x values and function values, and outputs the vector of coefficients of the interpolating polynomial.

3. The 2014 Kentucky Derby was won by a horse named California Chrome in a time of 2:03.66 (2 minutes and 3.66 seconds) for the $1\frac{1}{4}$ mile race. Times at the quarter-mile, half-mile, and mile poles were 0:23.04, 0:47.37, and 1:37.45.

- (a) Use these values together with the starting time to construct a natural cubic spline for California Chrome's race.
- (b) Use the spline to predict the time at the three-quarter-mile pole and compare this to the actual time of 1:11.80.
- (c) Use the spline to predict California Chrome's starting speed and speed at the finish.